6.3.6 The Fokker-Planck Equation

Consider now a collection of stochastic processes $\mathrm{X}(t)$ that are the solutions of the SDE $\mathrm{dX} = a(\mathrm{X})\mathrm{d}t + b(\mathrm{X})\mathrm{dW}$. If the initial conditions are drawn from a distribution with given density $\rho(x, 0)$ we would like to derive an equation for the corresponding density at time t, $\rho(x, t)$. We assume that ρ is C^∞ for all t and that the density tends to zero exponentially fast as $x \to \pm\infty$, whereas a and b are C^∞ functions which grow at most polynomially fast in these limits, that is, there are constants η_a, η_b such that

$$a(x) = O\left(|x|^{\eta_a}\right), \qquad b(x) = O\left(|x|^{\eta_b}\right), \qquad x \to \pm\infty.$$

This condition of polynomial growth of the coefficients is a common assumption in the analysis of SDEs. For future reference we define the space of functions which are C^∞ smooth and satisfy the polynomial growth restriction by C^∞_{P}:

$$C^\infty_{\mathrm{P}} = \left\{\phi \in C^\infty | \exists \eta \in \mathbb{N} : \phi(x) = \mathcal{O}(|x|^\eta), x \to \pm\infty\right\}.$$

For any $\phi \in C^\infty_{\mathrm{P}}$, the expectation with respect to $\rho(x, t)$ is well defined and we have

$$\mathbb{E}\phi(\mathrm{X}(t)) = \langle \phi, \rho \rangle = \int_{-\infty}^{+\infty} \phi(x)\rho(x, t)\mathrm{d}x.$$

Taking the differential, we obtain an equation for $\mathbb{E}\mathrm{d}\phi$ which must be equal to (6.37):

$$\int_{-\infty}^{+\infty} \phi(x)\frac{\partial \rho}{\partial t}\mathrm{d}x = \int_{-\infty}^{+\infty} (\mathcal{L}\phi)(x)\rho(x, t)\mathrm{d}x = \int_{-\infty}^{+\infty} (a(x)\phi' + \frac{1}{2}b^2(x)\phi''(x))\rho(x, t)\mathrm{d}x.$$

Using the assumptions given above regarding the exponential rate of decay of the density, it is possible to recast the right hand side as an integral of a product with ϕ. First, we find

$$\begin{aligned}\int_{-\infty}^{+\infty} a(x)\phi'(x)\rho(x, t)\mathrm{d}x &= [\phi(x)a(x)\rho(x, t)]|_{-\infty}^{\infty} - \int_{-\infty}^{+\infty} \phi(x)(a(x)\rho(x, t))_x\mathrm{d}x \\ &= -\int_{-\infty}^{+\infty} \phi(x)(a(x)\rho(x, t))_x\mathrm{d}x.\end{aligned}$$

With two rounds of integration by parts on the term involving $b(x)$, we find, using the fact that the boundary terms vanish,

$$\int_{-\infty}^{+\infty} b^2(x)\phi''(x)\rho(x,t)\mathrm{d}x = \int_{-\infty}^{+\infty} \phi(x)(b^2(x)\rho)_{xx}(x,t)\mathrm{d}x.$$

Finally, putting these results together, we have

$$\int_{-\infty}^{+\infty} \phi(x)\left[\frac{\partial\rho}{\partial t} + \frac{\partial}{\partial x}(a(x)\rho(x,t)) - \frac{1}{2}\frac{\partial^2}{\partial x^2}(b^2(x)\rho)\right]\mathrm{d}x = 0.$$

If this is to hold for all test functions ϕ, we must have

$$\frac{\partial\rho}{\partial t} = -\frac{\partial}{\partial x}(a(x)\rho) + \frac{1}{2}\frac{\partial^2}{\partial x^2}(b^2(x)\rho).$$

This partial differential equation is often called the Fokker-Planck equation. The great benefit of the Fokker-Planck equation is that it enables the macroscopic perspective on the solutions of an SDE.

The Fokker-Planck equation may be written in the compact form

$$\frac{\partial\rho}{\partial t} = \mathcal{L}^\dagger\rho,$$

where the operator appearing on the right hand side is the *adjoint* $\mathcal{L}^\dagger$ of the generator of the stochastic process, satisfying the relation

$$\langle\mathcal{L}\phi,\rho\rangle = \langle\phi,\mathcal{L}^\dagger\rho\rangle.$$

In the case at hand the adjoint is

$$\mathcal{L}^\dagger\rho = -(a(x)\rho)_x + \frac{1}{2}(b^2(x)\rho)_{xx},$$

which we shall refer to as the Kolmogorov operator.

In case $b \equiv 0$, the SDE becomes an ordinary differential equation. Then the generator $\mathcal{L}$ becomes simply the Lie derivative, and its adjoint is just the Liouvillian. Thus the framework developed here for SDEs is a proper generalization of what we have earlier seen in the deterministic case.